Lab: Using Calculated Fields for Advanced Analysis

Estimated time needed: **15** minutes.

In this lab, you will learn how to use calculated fields for advanced analysis in Tableau.

Required Software

You can complete this lab using the free, downloadable **Tableau Desktop (Public Edition)** desktop software. You may also use the free **Tableau Public online platform**, but some tasks may not work in exactly the same way or may not be supported in that version.

Prerequisites

You need a free Tableau Public account for this lab. If you haven't already done so, you can join up for a free Tableau Public account.

Data Set

This lab uses a sample data set called **World Indicators Data.xlsx** to showcase the Tableau features used in these lab exercises.

Hold the CTRL key while you click to download the following file:

World Indicators Data.xlsx

Objectives

The exercises in this lab teach you how to use calculated fields for advanced analysis in Tableau.

In this hands-on lab, you will:

1. Use calculated fields for advanced analysis

Exercise: Use calculated fields for advanced analysis

In this exercise, you will use the *World Indicators Data* dataset to illustrate some advanced dashboard capabilities in Tableau.

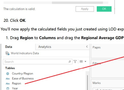
You'll use LOD expressions to find out Regional Average GDP per capita and Country Internet usage, and then you'll use one calculated field inside another calculated field, referred to as a nested calculated field, to calculate a complex formula for finding the %Y Health Expenditure Growth.

So, first, you'll calculate Regional Average GDP per Capita, Country Internet Usage, and %Y Health Expenditure Growth and then apply those to the visualization.

1. In the main menu, click **File**, and then click **Close**.
2. Under **To a File**, select **Text file**.
3. Navigate to the folder where you downloaded the data sets to, select **World Indicators Data.csv**, and click **Open**.
4. Click **Sheet 1** to extract the data.
5. In the main menu, click **Analysis**, and then click **Create Calculated Field**.
6. Name this field **Regional Average GDP per Capita**.
7. Enter the calculation formula **{FIXED [Region]: SUM([GDP]) / SUM ([Population Total])}**

8. Click **OK**.
9. In the main menu, click **Analysis**, and then click **Create Calculated Field**.
10. Name this new field **Country Internet Usage**.
11. Enter the calculation formula **{INCLUDE [Country/Region]: AVG ([Internet Usage])}**

12. Click **OK**.
13. In the main menu, click **Analysis**, and then click **Create Calculated Field**.
14. Name this new field **Previous Year Health Exp per Capita**.
15. Enter the calculation formula **LOOKUP (SUM([Health Exp/Capita]), -1)**

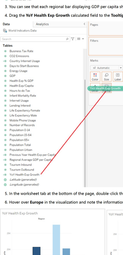


16. Click **OK**.
17. In the main menu, click **Analysis**, and then click **Create Calculated Field**.
18. Name this new field **%Y Health Exp Growth**.
19. Enter the calculation formula **(SUM ([Health Exp/Capita]) - [Previous Year Health Exp per Capita]) / ABS ([Previous Year Health Exp per Capita])**

20. Click **OK**.

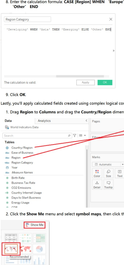
You'll now apply the calculated fields you just created using LOD expressions into a visualization.

1. Drag **Region** to **Columns** and drag the **Regional Average GDP per Capita** calculated field to **Rows**.



2. Drag the **Country Internet Usage** calculated field to the **Color** section on the **Marks** card.

3. Now, you can see that each regional bar displaying GDP per capita shows color variations according to the Internet Usage value.
4. Drag the **%Y Health Exp Growth** calculated field to the **Tooltip** section on the **Marks** card.



5. In the worksheet tab at the bottom of the page, double click the **Sheet 1** title, rename it to **%Y Health Exp Growth**, and press Enter.
6. Hover over **Europe** in the visualization and note the information in the Tooltip, then hover over **Oceania** and note the information in the Tooltip.

Next, you'll use complex logical conditions to show GDP Performance Tiers for categorizing Region.

So, first, you'll create a calculated field that uses an IF THEN ELSE statement, then create another one that uses the CASE statement, and then apply them to the visualization.

1. At the bottom of the page, click the **New Worksheet** icon.
2. In the main menu, click **Analysis**, and then click **Create Calculated Field**.
3. Name this field **GDP Performance Tier**.
4. Enter the calculation formula **IF SUM([GDP]) >= 1200000000000000 THEN 'High' ELSEIF SUM([GDP]) >= 200000000000000 THEN 'Medium' ELSE 'Low' END**

5. Click **OK**.
6. In the main menu, click **Analysis**, and then click **Create Calculated Field**.
7. Name this new field **Region Category**.
8. Enter the calculation formula **CASE [Region] WHEN 'Europe' THEN 'Developed' WHEN 'Africa' THEN 'Developing' WHEN 'Asia' THEN 'Emerging' ELSE 'Other' END**

9. Click **OK**.

Lastly, you'll apply calculated fields created using complex logical conditions into a comparative symbol map visualization.

1. Drag **Region** to **Columns** and drag the **Country/Region** dimension to **Rows**.

2. Click the **Show Me** menu and select **symbol maps**, then click the **Show Me** menu again to hide it.

3. In the worksheet tab at the bottom of the page, double click the **Sheet 2** title, rename it to **GDP Perf Tier**, and press Enter.

4. Drag the **GDP Performance Tier** calculated field to the **Columns** shelf.
5. Drag the **GDP Performance Tier** calculated field to the **Color** section on the **Marks** card.
6. Drag **Region Category** to the **Label** section on the **Marks** card.

7. Hover over the first map and click the + icon on the map tools once to zoom in on the maps.
8. Hover over **China** under the **Medium** column of the **Asia** map. The hover over **India** under the **Low** column of the **Asia** map.

9. This new visualization intricately segments and categorizes countries based on their GDP, providing a view of economic performance across different regions. The symbol map allows for a quick visual assessment of how countries stack up against each other within their respective tiers and categories.

Congratulations! You have completed this lab and are ready for the next topic.

Author(s)

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